

Product and Quotient Rules

Section 2.4

A. Product Rule

$$f(x) = g(x) \cdot h(x) \text{ then } f'(x) = g'(x) \cdot h(x) + g(x) \cdot h'(x)$$

Alternative Notation: $\frac{d(g \cdot h)}{dx} = \left(\frac{dg}{dx}\right) \cdot (h) + (g) \cdot \left(\frac{dh}{dx}\right)$

In Plain English: The derivative of the product of two functions (which we will call the “first” function and the “second” function) is equal to the **derivative of the first, times the second, plus the first, times derivative of the second.**

Examples

A. $f(x) = (2x^3 + 8x) \cdot (5x^4 + 17)$

We see that $f(x)$ consists of the product of two smaller functions, in this case $(2x^3 + 8x)$ “the first” and $(5x^4 + 17)$ “the second.” So, the derivative then is:

$$f'(x) = \underbrace{(6x^2 + 8)}_{\substack{\text{Derivative of} \\ \text{the first}}} \times \underbrace{(5x^4 + 17)}_{\text{The second}} + \underbrace{(2x^3 + 8x)}_{\text{The first}} \times \underbrace{(20x^3)}_{\substack{\text{Derivative of} \\ \text{the second}}}$$

Note: You should leave the answer in this form unless we are asked to “clean up.”
Again, do not forget to label your derivative.

B. $g(x) = x \cdot \sin x$

We see that $g(x)$ consists of the product of two smaller functions; in this case, x “the first” and $\sin x$ “the second.” So, the derivative then is: $g'(x) = (1) \cdot \sin x + x \cdot \cos x = \sin x + x \cdot \cos x$

More Examples: Find and LABEL the derivatives of each of the following functions.

1. **(video)** $f(x) = (3x^5 + 2\sqrt{x}) \cdot (4x^2 + 51)$

2. **(video)** $f(x) = 7x^2 \sin x$

3. $f(x) = (x^4 + \sqrt{x}) \cdot (5x - 1)$

4. $f(x) = x^2 \cos x$

5. $f(x) = \sin x \cos x$

B. Quotient Rule

$$f(x) = \frac{g(x)}{h(x)}, \text{ then } f'(x) = \frac{g'(x) \cdot h(x) - g(x) \cdot h'(x)}{[h(x)]^2}$$

Book Notation: $\frac{d\left(\frac{g}{h}\right)}{dx} = \frac{\left(\frac{dg}{dx}\right) \cdot (h) - (g) \cdot \left(\frac{dh}{dx}\right)}{h^2}$

In Plain English, the derivative of the quotient of two functions (which we will call the “top” function and the “bottom” function) is equal to the **derivative of the top, times the bottom, minus the top, times derivative of the bottom, all over the bottom squared.**

Examples

A. $f(x) = \frac{(2x^3 + 8x)}{(5x^4 + 17)}$

We see that $f(x)$ consists of the quotient of two smaller functions, in this case $(2x^3 + 8x)$ “the top” and $(5x^4 + 17)$ “the bottom.” So, the derivative then is:

$$f'(x) = \frac{\overbrace{(6x^2 + 8)}^{\text{Derivative of the top}} \times \overbrace{(5x^4 + 17)}^{\text{The bottom}} - \overbrace{(2x^3 + 8x)}^{\text{The top}} \times \overbrace{(20x^3)}^{\text{Derivative of the bottom}}}{\underbrace{(5x^4 + 17)^2}_{\text{The bottom squared}}}$$

Note: You should leave the answer in this form unless we are asked to “clean up.”

Again, do not forget to label your derivative.

B. $g(x) = \frac{x}{\sin x}$

We see that $g(x)$ consists of the product of two smaller functions, in this case x “the top” and $\sin x$ “the bottom.”

So, the derivative then is: $g'(x) = \frac{(1) \cdot \sin x - x \cdot \cos x}{[\sin x]^2} = \frac{\sin x - x \cdot \cos x}{\sin^2 x}$

More Examples: Find and LABEL the derivatives of each of the following functions:

6. (video) $f(x) = \frac{(12x^2 + 9\sqrt[3]{x})}{(5x^2 - 13)}$

7. (video) $f(x) = \frac{3x^5}{\cos x}$

8. $f(x) = \frac{(x^4 - 8x)}{(2x - 1)}$

9. $g(x) = \frac{\sin x}{\cos x}$

10. (video) $f(x) = \left(\frac{\sin x}{3x + 5} \right) \cdot (7x^2 + 2x)$

11. $l(x) = \frac{\sin x}{\sqrt{x}}(x^2 + 2x)$

12. Suppose $f\left(\frac{\pi}{6}\right) = 7$ and $f'\left(\frac{\pi}{6}\right) = -5$, and let $g(x) = f(x)\cos x$ and $h(x) = \frac{\sin x}{f(x)}$

$g'\left(\frac{\pi}{6}\right) =$

$h'\left(\frac{\pi}{6}\right) =$

13. Use the table to find the value of the following

x	1	0	7	-2	-1
$f(x)$	-5	-1	-407	7	1
$g(x)$	-3	-2	-9	0	-1
$f'(x)$	-7	-2	-163	-10	-3
$g'(x)$	-1	-1	-1	-1	-1

a. $(fg)'(-1) =$

b. $\frac{f(-1)}{g(-1)+5}$

c. $(f+g)'(-1) =$

d. $(f-g)'(-1) =$

e. $(fg)'(7) =$

f. $\left(\frac{g}{f}\right)'(0) =$

14. Given that

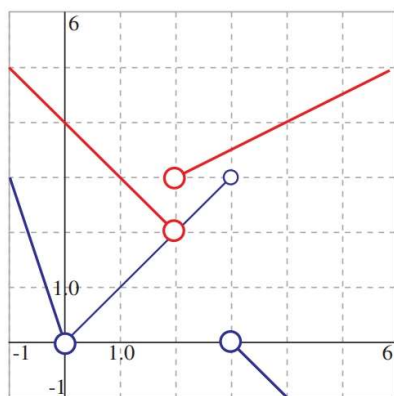
$$f(x) = x^{12}h(x)$$

$$h(-1) = 3$$

$$h'(-1) = 6$$

Calculate $f'(-1)$

15. The graphs of the function f (given in blue) and g (given in red) are plotted below. Suppose that $u(x) = f(x) \cdot g(x)$ and $v(x) = \frac{f(x)}{g(x)}$.



f is the bottom function

g is the top function

a. Find $u'(1)$

b. Find $v'(1)$